



Project
Management
Learning Hub



Skill Development
Council Canada



INTERNATIONAL ASSOCIATION OF
PROJECT MANAGERS



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PROJECT MANAGERS

CECOS
UNIVERSITY

Professional Certificate in Project Planning and Scheduling (Course Curriculum)

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Lahore, Pakistan



Course Pre-requisite	12-14 Years of Education No Experience Required
Course Duration	02 Months (Weekend + Regular Classes) 135 Credit Hours
Instructor Name	Prof. Dr. Faisal Fayyaz

1. Course Objective

The main objective of the course is to provide the skills and knowledge you need to successfully launch, plan, estimate, and schedule your project. Work in project teams to create planning related documents while going through the planning process. The critical path method (CPM) basics are covered in this course, along with how to create schedules based on the results of a CPM network analysis. For engineers and project managers to effectively plan, manage, and finish their projects on schedule, CPM scheduling is a crucial ability.

It is a dedicated two-month training module in a structured and professional manner allow candidates to demonstrate following knowledge and skills:

- Discover step-by-step how to plan and schedule your Project.
- Use the principles of network analysis.
- Create schedules based on the results of the CPM network study.
- Apply the activity sequencing concepts to common tasks of all kinds.
- Describe how resource levelling is used to improve the planned order of operations.
- Explain the cash flow scheduling method that uses cost loading.
- To simplify planning tactics, shorten overall timetables, and save money, identify and prioritize scheduled jobs.
- Recognize and apply best practices for organizing and scheduling project tasks to effectively communicate with team members.
- Apply scheduling tools to manage and analyze project schedules
- Understand the importance of the critical path in the determination of the project's completion

Enrolling in this course will assist you in comprehending project scheduling basics and advance features and functionality so that you can apply them to your company's project management operations. You'll learn how to plan and schedule projects on time and on budget in order to get the most out of your capital and resource expenditures.

By taking this course, you will have a better understanding of project scheduling techniques and functionality allowing you to make better project management decisions. You'll learn how to efficiently manage resources in order to maximize investment, as well as how to choose and prioritize projects in order to align strategy, execution, and results.

2. Course Delivery

- a) Classroom delivery at PMLH classroom

- b) Online delivery by using “Zoom” application with real life project examples and hand on practice on MS Project.

3. **Course Curriculum**

Lecture Day	Topic	Detail
1	Introduction to Project Management	• What is project? • Why are Projects important? • Project Life Cycles • Project Elements
2	Project Scheduling	• Project Scheduling Introduction • Project scheduling techniques
3	Developing a network	• Labeling Nodes • Serial activities • Concurrent activities • Merge activities • Burst activities
4	Duration Estimation	• Duration Estimation Introduction • Duration Estimation Techniques • PERT • Normal distribution
5	Constructing Critical Path	• Network diagram • Forward pass • Backward pass • Laddering activities • Hammock activities • Steps to reduce critical path
6	Project Lags	• Relationship types • Lag types
7	Gantt Chart	• Create Gantt Chart • Add resources • Incorporate lags
8	Crashing Projects	• Accelerating the activities • Crashing the project
9	Activities	• Activities on arrow • Activities on node • Dummy activities
10	Constraints	• Constraint types • Common cause of variation • Special cause of variation
11	Critical Chain Path Method	• Develop critical chain network • Critical chain method • Critical Path method • Conflict resolution
12	Scheduling Model	• Baseline • Milestones

		• Activities • Resources • Duration • Analyze schedule • Approval of schedule
13	Schedule Model Maintenance	• Actual and remaining work • Update schedule • Compare and deviation • Approved changes • Update baseline • Record maintenance
14	Schedule model analysis	• Total float and free float • Estimation • Date constraints • Out of sequence logics • Risk assessment • Earned Schedule
15	Conformance Index	• Categories • Components • Assessment Process • Assessment
16	Project Planning	• Project planning Introduction • Planning techniques
17	Project submission / Examination	• Final Project Submission • MCQ examination (one hour duration)

4. Expected Audience

This course is appropriate for anyone who is a member of a team that works on industrial projects. If at all possible, groups will be made up of organizations that complement each other in order for all participants to get as much knowledge as possible.

It is also highly useful for planning and scheduling the tasks, resources, risks, issues and controlling the budget, assessing deadlines, analyzing resource needs, and monitoring project progress. Our goal is to educate you on how to plan and schedule a project in the most effective way possible. Our course will teach you the planning best practices, and you will be ready to plan and schedule your projects easily. This Certification Course will give you practical experience that will help you advance in your career.

The typical users are:

- Project Managers
- Engineers and Planners
- Trades People wanting to upskill
- Contracts and Claims Support
- Support Engineer

- Analyst
- End User
- Implementer
- Manager
- Fresher's or early-career professionals
- Schedulers
- Project Controls Engineer

5. Course Outcome

- Better planning
- Better scheduling with accurate duration estimate
- Improve planning and tracking skills,
- Enhance project management skills.
- Increase efficiency.
- Professional presence
- Better organize trades and materials

6. Grading System

Grade	Percentage	Performance
A	90-100%	Excellent
B	80-89%	Good
C	70-79%	Average
D	65-69%	Satisfactory
F	0-65%	Fail